

MEMformer: Transformer-based 3D Human Motion Estimation from MoCap Markers

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Figure 1: Applications of our proposed method. From left to right: the input optical markers, the output body mesh, estimated joint positions, the body mesh retargeted to a different subject, and garment simulation on body mesh.

ABSTRACT

We address the problem of 3D human motion estimation from original MoCap optical markers. The original markers are noisy, disordered, and unlabeled, hence recovering 3D human motion from them is non-trivial. Existing works are either time-consuming or assuming the knowledge of the marker labels. We address these problems by presenting an end-to-end method for 3D human motion estimation by leveraging the capability of Transformer to model long-range dependencies. The method takes original markers as inputs and learns joint poses with a Transformer-like architecture. Experimental results show that our method is able to achieve better than centimeter-level errors.

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1 INTRODUCTION

With the rise of the Metaverse in recent years, we have witnessed an explosion of interests in 3D digital humans because of its wide applications in 3D virtual contents, such as video games, 3D animations, and film industries. All these applications require to create 3D

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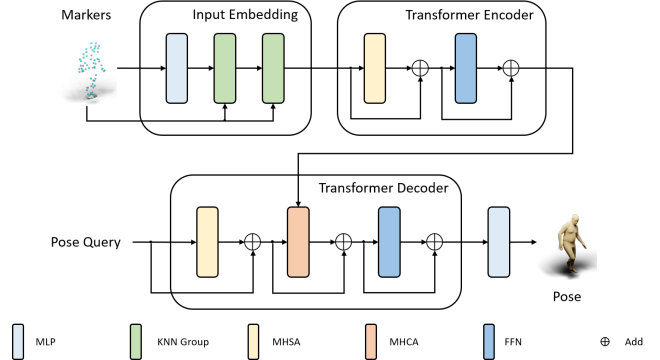


Figure 2: The overview of our proposed method. **KNN Group** concatenates the K-nearest neighbor of each marker to learn neighbor contexts. **MHSA** represents a multi-head self-attention module and **MHCA** denotes a multi-head cross-attention module. **Symbol $\oplus$** calculates the element-wise addition.

digital humans with vivid and diverse motions. Existing commercial solutions obtain human motions by real-time motion capture technologies, where a series of motions are performed by professional actors with markers placed on their bodies according to the MoCap protocol. Then trajectories of markers can be obtained through specialized sensors and further converted to a series of skeletons. However, captured markers are usually accompanied with noise and occlusion. The industry usually hinders these problems by expensive equipment and tedious subject calibrations, making motion capture inaccessible to general users. Therefore, it is important to ease the usage of optical markers.

Traditional methods such as MoSh++ [Mahmood et al. 2019] take an iterative approach to optimize the differences between simulated markers transformed from the t-posed shape and observed markers

Table 1: Quantitative results and ablation studies.

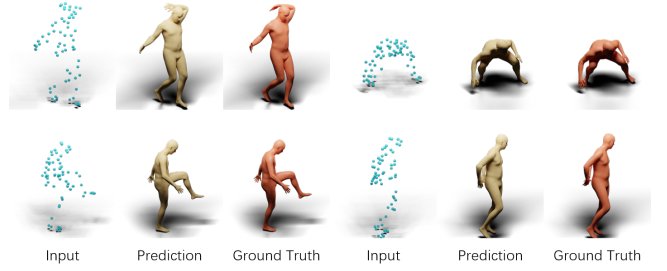
Methods	MPJPE (cm)	MPVPE (cm)
MoSh++ [Mahmood et al. 2019]	3.46	3.59
Standard Transformer	5.76	7.11
Ours-MLPEmb	1.37	1.80
Ours-MSE $\mathcal{L}_d$	1.10	1.36
Ours-w/o $\mathcal{L}_j$	1.02	1.19
Ours-w/o $\mathcal{L}_v$	0.99	1.90
Ours-w/o $\mathcal{L}_j$ & $\mathcal{L}_v$	3.88	4.52
Ours	0.87	1.20
Ours-(Double Training Data)	0.81	1.03
Ours-(Half Training Data)	1.57	2.05
Ours-(Quarter Training Data)	1.72	2.29

of a posed shape based on Powell’s Dogleg method. But these energy optimization methods are usually time-consuming and rely on the consistency of markers. DeepMurf [Madadi et al. 2021] firstly predicts the missing markers and joints, then regresses the parameters of SMPL models [Loper et al. 2015] with a cascading network. However, all above methods require semantic labels of MoCap markers to provide a prior of the relative position relationship between markers and human body. In order to address the dependency on marker labels, SOMA [Ghorbani and Black 2021] predicts the label assignment for each marker with attention mechanism. Nevertheless, it needs further post-processing by MoSh++ [Mahmood et al. 2019] and is not an end-to-end solution.

In this work, we address above problems by presenting the first end-to-end method to estimate 3D human motions from original MoCap optical markers. The method does not require marker labels and has no constraint on the layout of markers. Considering the long-range dependency between joint poses, we propose a Transformer-based network to tackle the problem. First, we employ input embedding module to learn local features of markers. Then, we encode the relation between any two markers or joints with a self-attention module and the relation between joint poses and marker features with a cross-attention module. In this way we can recover 3D human motion from MoCap markers without any requirement on markers or manual intervention.

2 OUR APPROACH

Our network consists of an input embedding module, a Transformer encoder module, and a Transformer decoder module, as shown in Fig. 2. The input embedding module learns local point-wise features using a multi-layer perceptron (MLP) and two K-nearest neighbor (KNN) group modules. Then point-wise features are further enhanced by a Transformer encoder. We employ a multi-head self-attention module (MHSA) to learn the long-range dependency of human poses and concentrate more on upper joints in the SMPL kinematic tree. Last but not least, we employ a multi-head cross-attention module (MHCA) to learn the global dependency between output joint poses and input optical markers. To train the network, we minimize the joint position loss $\mathcal{L}_j$ and vertex position loss $\mathcal{L}_v$ with $l1$ norm and the weighted mean-squared error (MSE) loss $\mathcal{L}_d$ of predicted pose parameters, which could also help learn the long-range dependency of joint poses.

**Figure 3: Qualitative results.**

3 EXPERIMENTS AND RESULTS

We first synthesize sequences of markers and then add Gaussian noises to simulate unavoidable noises on markers. Networks are trained on SURREAL dataset, and experiments are evaluated with mean per-joint position errors (MPJPE) and mean per-vertex position errors (MPVPE). The quantitative results are shown in Tab. 1. We compare our method with MoSh++ [Mahmood et al. 2019] which is not real-time and takes 5,000 frames labeled markers as inputs. For the reason that it outputs SMPL-X parameters, we transfer the results to SMPL parameters for comparison. In ablation study, we first show that a naïve Transformer-based network produces very bad performance. This is probably because a masked self-attention module in decoder ignores the long-range dependency of joint poses and the post-normalization leads to an unstable training process. Then we validate different designs of the proposed method. Finally, we show the impact of different quantities of training data. Even though more training data makes the results more accurate, it also leads to more computational expense. The visualization results are shown in Fig. 3, which indicates that the errors are obvious in the limbs and head parts. The reason is that SMPL shapes use fewer joints to represent the motions of these parts.

4 CONCLUSION

In this work, we address the problem of human motion estimation from original MoCap optical markers. We propose a Transformer-based framework to learn long-range dependency between output joint poses and input markers for an end-to-end human motion recovery. Experiments on synthesized dataset show the proposed method is effective.

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